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Supercritical in-mold foam forming device with pretreatment function and foaming method thereof

Abstract

Disclosed are a supercritical in-mold foam forming device with a pretreatment function and a foaming method thereof. The supercritical in-mold foam forming device comprises a high-pressure reactor. A safety valve is disposed at the top of the high-pressure reactor and seals the top of the high-pressure reactor. A driving member used for driving is mounted in the middle of the top of the high-pressure reactor through bolts. An intra-mold temperature control module is disposed in a foaming mold, two groups of uniform heating plates are controlled be powered on synchronously to generate heat, and a temperature gradient is formed in a graphite backing plate by controlling the difference in proportion of output power.

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Background/Summary

BACKGROUND OF THE PRESENT INVENTION

(1) The invention relates to supercritical foaming-related fields, in particular to a supercritical in-mold foam forming device with a pretreatment function and a foaming method thereof.

DESCRIPTION OF RELATED ARTS

(2) During the foam forming or polymer material foaming process, a physical foaming agent or chemical foaming agent is added for reaction to form a cellular structure or a porous structure. The foam forming process comprises the basic steps of bubble nucleation, bubble growth or expansion,

and bubble stabilization; and under given temperature and pressure conditions, gas reaches a saturated state with the decrease of its solubility, and redundant gas is discharged and bubbles are formed to realize nucleation.

(3) An existing compression molding method for preparing microcellular foamed plastic cannot sufficiently mix supercritical gas and polymer and cannot realize uniform distribution of cells of a prepared material, thus limiting further improvement of the performance of the material.

(4) In addition, most existing pretreatment methods perform pretreatment merely by stirring and mixing, which lead to low mixing efficiency and makes it difficult to quickly infiltrate a foaming agent into foam particles, so the foam material has to stand in the foaming agent for an excessively long time, compromising the foaming efficiency.

SUMMARY OF THE PRESENT INVENTION

(5) To overcome the abovementioned defects, the invention provides a supercritical in-mold foam forming device with a pretreatment function and a foaming method thereof.

(6) The invention is implemented as follows: a supercritical in-mold foam forming device with a pretreatment function and a foaming method thereof are provided. The supercritical in-mold foam forming device comprises a high-pressure reactor, a safety valve is disposed at the top of the high-pressure reactor and seals the top of the high-pressure reactor, a driving member used for driving is mounted in the middle of the top of the high-pressure reactor through bolts, a foaming mold is mounted on a support frame at the bottom of the high-pressure reactor through bolts, and a control panel is mounted in the middle of a right end of high-pressure reactor through bolts;

(7) Wherein, the supercritical in-mold foam forming device further comprises a pretreatment module disposed at a front end of the high-pressure reactor;

(8) The pretreatment module comprises: a heating box used for heating an inert gas to control the temperature of the inert gas and mounted at a front end of the high-pressure reactor through bolts; a one-way metering valve fixedly mounted at a pipe opening connected to a left end of the heating box; a programmable control box mounted at the top of the heating box through bolts, a heat-insulation cushion being disposed between the heating box and the programmable control box; an intra-mold temperature control module disposed in the foaming mold and connected to the programmable control box through a cable; a temperature control valve mounted at a through hole in the bottom of a right end of the heating box through a pipe; a radial gas cutting module disposed in the high-pressure reactor; and an axial stirring module disposed at the bottom of the driving member; wherein, the driving member is specifically composed of a gearbox and a motor, and the one-way metering valve and the temperature control valve are connected to interfaces of the programmable control box through cables.

(9) Preferably, the intra-mold temperature control module comprises: matching mold bodies fixedly mounted on opposite upper and lower sides of the foaming mold and in seal fit the upper and lower sides of the foaming mold; uniform heating plates mounted on inner sides of the matching mold bodies on the upper and lower sides of the foaming mold through bolts; an intermediate mold body, the matching mold bodies being in seal fit with upper and lower side of the intermediate mold body; a graphite backing plate used for heat conduction and disposed on an inner wall of the intermediate mold body; a high-sensitivity temperature gauge disposed in the intermediate mold body; a data switching unit connected to and mounted at the bottom of the high-sensitivity temperature gauge through a cable; and a horizontal connecting plate inserted and fixed in a front side of the intermediate mold body.

(10) Preferably, a heat-conducting material layer is filled and fixed in the inner wall of the intermediate mold body, and the high-sensitivity temperature gauge is fixed in an internal space of the heat-conducting material layer.

(11) Preferably, the radial gas cutting module comprises: a diverter valve, a right end of the temperature control valve being connected to a gas port in the bottom of the diverter valve through a pipe; a retainer ring rotatably disposed at the top of the diverter valve; rigid tubes used for flow

guiding and inserted and fixed in through holes in an outer circular side of the diverter valve; a hollow ring, the rigid tubes being connected to through holes in an inner circular wall of the hollow ring through pipes; a ceramic plate, the hollow ring being connected to upper and lower sides of the ceramic plate through bolts respectively; a sprayer disposed on an inner circular wall of the ceramic plate; and heat-resistant ring tubes, through holes in the upper and lower sides of the ceramic plate being connected to a gas outlet of the hollow ring through heat-resistant ring tubes respectively.

(12) Preferably, a central point of the retainer ring and a center of the hollow ring are located at a same height, and six groups of rigid tubes are disposed on the outer side of the diverter valve, and outer sides of the rigid tubes are coated with heat-insulation coatings.

(13) Preferably, the sprayer comprises: a spraying plate mounted on an inner wall of the ceramic plate through bolts; a through groove formed in the spraying plate; a conical assembly fixedly mounted in the through groove; a polymer heat-resistant membrane filled in a space between the conical assembly and the through groove; an elastic element disposed on a conical head of the conical assembly; and a sealing plug, an outer end of the elastic element being elastically connected to the sealing plug.

(14) Preferably, a gas hole is formed in a position, communicated with the through groove, of the conical assembly, the through groove and the conical assembly are identical in shape and are distributed in a proportion of 1:1.2, strip-shaped gas holes and strip-shaped lugs matched with the strip-shaped gas holes are disposed on the sealing plug and a through hole in a rear end of the through groove, and the polymer heat-resistant membrane is specifically a water-proof and breathable membrane.

(15) Preferably, the axial stirring module comprises: a driven shaft inserted and fixed in an inner side of the retainer ring; a planetary gear set, a top of the driven shaft being inserted and fixed in a center of a sun gear of the planetary gear set, the planetary gear set being specifically composed of the sun gear, a planetary gear and a gear ring, the planetary gear of the planetary gear set being inserted and fixed in a shaft of the gearbox of the driving member, and a gear ring plate of the planetary gear set being fixed to an inner wall of the hollow ring; axial blades inserted and fixed to upper and lower sides of an outer shaft of the driven shaft; a grinding block used for grinding and mounted on an outer side of a middle section of the driven shaft; a grinding ring disposed on an outer side of the grinding block; a lantern ring, the grinding ring being mounted on an inner wall of the lantern ring through bolts; and a heat-insulation cross rod, an outer side of the lantern ring being fixedly mounted on an inner circular surface of the spraying plate through the heat-insulation cross rod.

(16) Preferably, opposite upper and lower sides of the grinding block and the grinding ring are arc-shaped, and a distance between the grinding block and the grinding ring is 1 cm.

(17) Preferably, the foaming method of the supercritical in-mold foam forming device with a pretreatment function comprises:

(18) Step 1: first, connecting, by a worker, an external pipe to the one-way metering valve, and guiding an inert gas into the heating box to be heated, wherein the inert gas is specifically carbon dioxide commonly used during a foaming process; then, guiding the heated inert gas into the radial gas cutting module through the temperature control valve to sweep and preheat an interior of the high-pressure reactor, then enabling the inert gas to flow back into the heating box through a pipe at the top of the high-pressure reactor, and opening the safety valve to add a foam material and a foaming agent into the high-pressure reactor;

(19) Step 2: controlling, through the control panel, the driving member to drive the planetary gear set to drive the driven shaft and the hollow ring to rotate in opposite directions, at this moment, driving, by the axial blades, the foam material and the foaming agent to flow from bottom to top to enter the hollow ring, enabling the grinding block and the grinding ring to rotate in opposite directions to grind the foam material, and then, guiding the inert gas into the radial gas cutting module and spraying it out through the through hole in the inner side of the through groove, such

that the foam material and the foaming agent are sheared and stirred in a radial direction by the radial gas cutting module, and are stirred and ground in an axial direction by the axial blades and the grinding block (474), making the foaming agent infiltrate into the foam material more quickly; (20) Step 3: controlling, by means of the programmable control box, the two groups of uniform heating plates to be powered on synchronously to generate heat, at this moment, controlling power output to the two groups of uniform heating plates to be different, wherein input power of the uniform heating plate at the top is maximum, and input power of the uniform heating plates below is decreased in proportion; a difference in output heat of the uniform heating plates leads to a temperature gradient in the graphite backing plate, the heat is reflected around the high-sensitivity temperature gauge in equal proportion through filler materials in the intermediate mold body, and the proportion of the input power of the uniform heating plates is calculated by adjusting a heat value gradient detected by the high-sensitivity temperature gauge according to a preset value in the programmable control box; and

(21) Step 4: discharging a foamed preform out of the foaming mold through a discharge unit which is a common device used during the foaming process such as an extruder, then enabling the foamed preform to enter the graphite backing plate, making a 3D foaming ratio of the 3D preform nonsynchronous under the influence of the temperature gradient, with a foaming ratio along a length and width in a horizontal direction being small and a foaming ratio in a vertical direction being large, and then performing compression molding, wherein to guarantee the nonsynchronous 3D foaming ratio, a thickness of an internal space of the graphite backing plate is kept at a value which ensures that the heat output efficiency of the uniform heating plates, the heat conductivity of the graphite backing plate and the foam material, and the thickness of the internal space of the graphite backing plate are in direct proportion.

(22) The invention has the following advantages: the invention provides, through improvement, a supercritical in-mold foam forming device with a pretreatment function and a foaming method thereof, which compared with similar devices, are improved in the following aspects:

(23) According to the supercritical in-mold foam forming device with a pretreatment function and the foaming method thereof provided by the invention, the pretreatment module is disposed at the front end of the high-pressure reactor, an inert gas is guided, through the one-way metering valve, into the heating box to be heated, then the heated inert gas is guided into the radial cutting module through the temperature control valve, the axial stirring module is driven by the driving member, and the foam material and the foaming agent are sheared and stirred in radial direction by the radial gas cutting module and are stirred and ground in the axial direction by the axial stirring module, such that the foaming agent can infiltrate into the foam material more quickly.

(24) According to the supercritical in-mold foam forming device with a pretreatment function and the foaming method thereof provided by the invention, the intra-mold temperature control module is disposed in the foaming mold, the two groups of uniform heating plates are controlled to be powered on synchronously to generate heat, a temperature gradient is formed in the graphite backing plate by controlling the difference in proportion of output power, heat is reflected around the high-sensitivity temperature gauge in equal proportion through the filler materials in the intermediate mold body, the foamed preform is conveyed into the graphite backing plate, the 3D foaming ratio of the 3D preform is nonsynchronous under the influence of the temperature gradient, with the foaming ratio along the length and width in the horizontal direction being small and the foaming ratio in the vertical direction being large, and then compression molding is performed.

(25) According to the supercritical in-mold foam forming device with a pretreatment function and the foaming method thereof provided by the invention, the radial gas cutting module is disposed in the high-pressure reactor, the gas flow from the temperature control valve enters the ceramic plate through the rigid tubes and the hollow ring by means of the diverter valve, and at this moment, the gas pressure in the ceramic plate is increased to push the sealing plug to squeeze the elastic

element, such that the sealing plug horizontally moves in the through groove; at this moment, the gas flow enters the conical assembly through the gap and is guided into the high-pressure reactor through the polymer heat-resistant membrane, the sealing plug is pushed by the gas pressure to move, the through groove is communicated with the interior of the ceramic plate through the strip-shaped groove formed in the outer side of the sealing plug, the space between the through groove and the interior of the ceramic plate can be quickly sealed by means of the elastic potential energy of the elastic element when the gas pressure decreases, and the foaming agent is prevented from flowing into the through groove by means of the polymer heat-resistant membrane which is made of a water-proof and breathable material, so the hollow ring is prevented from being blocked, the situation where the proportion of materials is out of balance due to the loss of the foaming agent is avoided, the influence on foaming infiltration is reduced, and the interior of the high-pressure reactor is better swept and preheated.

(26) According to the supercritical in-mold foam forming device with a pretreatment function and the foaming method thereof provided by the invention, the axial stirring module is disposed in the high-pressure reactor, the driving member drives the planetary gear set to drive the driven shaft and the hollow ring to rotate in opposite directions, and at this moment, the axial blades drive the foam material and the foaming agent to flow axially in a direction parallel to the driven shaft, such that the foam material and the foaming agent flow from bottom to top to enter the hollow ring; the grinding block and the grinding ring rotate in opposite directions to grind the foam material, and then an inert gas is guided into the radial gas cutting assembly to shear and stir the foam material and the foaming agent in the radial direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a structural view of the invention;
- (2) FIG. 2 is a three-dimensional sectional structural view of a high-pressure reactor and a pretreatment module of the invention;
- (3) FIG. 3 is a three-dimensional structural view of a radial gas cutting module of the invention;
- (4) FIG. 4 is an enlarged structural view of part A in FIG. 2 of the invention;
- (5) FIG. 5 is a three-dimensional structural view of the radial gas cutting module and an axial stirring module of the invention;
- (6) FIG. 6 is a three-dimensional structural view of the axial stirring module of the invention;
- (7) FIG. 7 is a structural view of a grinding block and a grinding ring of the invention;
- (8) FIG. 8 is a right structural view of a foaming mold and an intra-mold temperature control module of the invention.
- (9) Reference signs: 1, high-pressure reactor; 2, safety valve; 3, driving member; 4, pretreatment module; 5, foaming mold; 6, control panel; 41, heating box; 42, one-way metering valve; 43, programmable control box; 44, intra-mold temperature control module; 45, temperature control valve; 46, radial gas cutting module; 47, axial stirring module; 441, matching mold bodies; 442, uniform heating plate; 443, intermediate mold body; 444, graphite backing plate; 445, high-sensitivity temperature gauge; 446, data switching unit; 447, horizontal connecting plate; 461, diverter valve; 462, retainer ring; 463, rigid tube; 464, hollow ring; 465, ceramic plate; 466, sprayer; 467, heat-resistant ring tube; 4661, spraying plate; 4662, through groove; 4663, conical assembly; 4664, polymer heat-resistant membrane; 4665, elastic element; 4666, sealing plug; 471, driven shaft; 472, planetary gear set; 473, axial blade; 474, grinding block; 475, grinding ring; 476, lantern ring; 477, heat-insulation cross rod.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

- (10) The principle and features of the invention will be described below in conjunction with FIG.

1-FIG. 8. Examples provided below are used for explaining the invention rather than limiting the scope of the invention. The advantages and features of the invention will be clearer with reference to the following description and claims. It should be noted that the drawings are in a simplified form and are not in accurate scale and they are merely used to assist in explaining the purposes of the embodiments of the invention easily and clearly.

(11) It should be pointed out that when one module is referred to as being “fixed” the other module, it may be directly fixed to the other module, or there may be another module between the two modules. When one module is referred to as being “connected” to the other module, it may be directly connected to the other module, or there may be another module between the two modules. When one module is referred to as being “disposed” on the other module, it may be directly disposed on the other module, or there may be another module between the two modules. Terms such as “vertical”, “horizontal”, “left” and “right” and similar expressions are merely for a purpose of description.

(12) Unless otherwise defined, all technical and scientific terms used in this specification have meanings as commonly understood by those skilled in the art. Terms used in the specification of the invention are merely used to describe specific embodiments, and are not intended to limit the invention. The term “and/or” used in this specification refers to any one or all combinations of one or more relevant items listed.

(13) Referring to FIG. 1-FIG. 8, the invention provides a supercritical in-mold foam forming device with a pretreatment function and a foaming method thereof. The supercritical in-mold foam forming device comprises a high-pressure reactor **1**, wherein a safety valve **2** is disposed at the top of the high-pressure reactor **1** and seals the top of the high-pressure reactor **1**, a driving member **3** used for driving is mounted in the middle of the top of the high-pressure reactor **1** through bolts, a foaming mold **5** is mounted on a support frame at the bottom of the high-pressure reactor **1** through bolts, and a control panel **6** is mounted in the middle of a right end of high-pressure reactor **1** through bolts;

(14) Referring to FIG. 1-FIG. 2, according to the supercritical in-mold foam forming device with a pretreatment function and the foaming method thereof provided by the invention, the pretreatment module **4** comprises a heating box **41**, the heating box **41** used for heating an inert gas to control the temperature of the inert gas is mounted at a front end of the high-pressure reactor **1** through bolts, a one-way metering valve **42** is fixedly mounted at a pipe opening connected to a left end of the heating box **41** and is used for metering the flow rate of the inert gas and guiding the inert gas, a programmable control box **43** is mounted at the top of the heating box **41** through bolts, a heat-insulation cushion is disposed between the heating box **41** and the programmable control box **43**, the programmable control box **43** is connected to an intra-mold temperature control module **44** in the foaming mold **5** through a cable, all components of the intra-mold temperature control module **44** are controlled by means of a program input to the programmable control box **43**, a temperature control valve **45** is mounted at a through hole in the bottom of a right end of the heating box **41** through a pipe, a radial gas cutting module **46** is disposed in the high-pressure reactor **1**, an axial stirring module **47** is disposed at the bottom of the driving member **3**, the driving member **3** is specifically composed of a gearbox and a motor, and the one-way metering valve **42** and the temperature control valve **45** are connected to interfaces of the programmable control box **43** through cables and are controlled by means of the programmable control box **43**.

(15) Referring to FIG. 8 and FIG. 2, according to the supercritical in-mold foam forming device with a pretreatment function and the foaming method thereof, the intra-mold temperature control module **44** comprises matching mold bodies **441**, the matching mold bodies **441** are fixedly mounted on opposite upper and lower sides of the foaming mold **5** and are in seal fit the upper and lower sides of the foaming mold **5**, uniform heating plates **442** are mounted on inner sides of the matching mold bodies **441** on the upper and lower sides of the foaming mold **5** through bolts, and are installed and limited by means of the matching mold bodies **441**, the matching mold bodies **441**

are in seal fit with upper and lower side of an intermediate mold body **443**, a graphite backing plate **444** used for heat conduction is disposed on an inner wall of the intermediate mold body **443**, heat in the intermediate mold body **443** can be conducted by means of the high heat conductivity of the graphite backing plate **444**, a high-sensitivity temperature gauge **445** is disposed in the intermediate mold body **443**, a data switching unit **446** is connected to and mounted at the bottom of the high-sensitivity temperature gauge **445** through a cable, a data switching effect is provided for the high-sensitivity temperature gauge **445** by means of the data switching unit **446**, a horizontal connecting plate **447** is inserted and fixed in a front side of the intermediate mold body **443**, a heat-conducting material layer is filled and fixed in the inner wall of the intermediate mold body **443**, and the high-sensitivity temperature gauge **445** is fixed in an internal space of the heat-conducting material layer.

(16) Referring to FIG. 3 and FIG. 2, according to the supercritical in-mold foam forming device with a pretreatment function and the foaming method thereof provided by the invention, the radial gas cutting module **46** comprises a diverter valve **461**, a right end of the temperature control valve **45** is connected to a gas port in the bottom of the diverter valve **461** through a pipe, a retainer ring **462** is rotatably disposed at the top of the diverter valve **461**, the diverter valve **461** is installed and limited by means of the retainer ring **462**, rigid tubes **463** used for flow guiding are inserted and fixed in through holes in an outer circular side of the diverter valve **461** and are connected to through holes in an inner circular wall of a hollow ring **464** through pipes, the hollow ring **464** is connected to upper and lower sides of a ceramic plate **465** through bolts, a gas flow is guided into the ceramic plate **465** through the hollow ring **464**, a sprayer **466** is disposed on an inner circular wall of the ceramic plate **465**, through holes in the upper and lower sides of the ceramic plate **465** are connected to a gas outlet of the hollow ring **464** through heat-resistant ring tubes **467** respectively, the central point of the retainer ring **462** and the center of the hollow ring **464** are located at the same height, six groups of rigid tubes **463** are disposed on the outer side of the diverter valve **461**, outer sides of the rigid tubes **463** are coated with heat-insulation coatings, and heat dissipation of the gas flow in the rigid tubes **463** is reduced by means of the heat insulating property of the rigid tubes **463**.

(17) Referring to FIG. 4, according to the supercritical in-mold foam forming device with a pretreatment function and the foaming method thereof provided by the invention, the sprayer **466** comprises a spraying plate **4661** mounted on an inner wall of the ceramic plate **465** through bolts, a through groove **4662** is formed in the spraying plate **4661**, a conical assembly **4663** is fixedly mounted in the through groove **4662** and provides a flow resisting effect for the through groove **4662**, the space between the conical assembly **4663** and the through groove **4662** is filled with a polymer heat-resistant membrane **4664** which makes the through groove **4662** water-roof, heat-proof and breathable, an elastic element **4665** is disposed on a conical head of the conical assembly **4663**, an outer end of the elastic element **4665** is elastically connected to a sealing plug **4666**, a gas hole is formed in a position, communicated with the through groove **4662**, of the conical assembly **4663**, the through groove **4662** and the conical assembly **4663** are identical in shape and are distributed in a proportion of 1:1.2, strip-shaped gas holes and strip-shaped lugs matched with the strip-shaped gas holes are disposed on the scaling plug **4666** and a through hole in a rear end of the through groove **4662**, the gas flow is guided by means of the strip-shaped gas holes of the sealing plug **4666** and the through groove **4662**, and the polymer heat-resistant membrane **4664** is specifically a water-proof and breathable membrane.

(18) Referring to FIG. 5-FIG. 7, according to the according to the supercritical in-mold foam forming device with a pretreatment function and the foaming method thereof provided by the invention, the axial stirring module **47** comprises a driven shaft **471** inserted and fixed in an inner side of the retainer ring **462**, the top of the driven shaft **471** is inserted and fixed in the center of a sun gear of a planetary gear set **472**, the planetary gear set **472** is specifically composed of the sun gear, a planetary gear and a gear ring, the planetary gear of the planetary gear set **472** is inserted

and fixed in a shaft of the gearbox of the driving member **3**, a gear ring plate of the planetary gear set **472** is fixed to an inner wall of the hollow ring **464**, power is supplied to the driven shaft **471** and the hollow ring **464** by means of the planetary gear set **472**, axial blades **473** are inserted and fixed in upper and lower sides of an outer shaft of the driven shaft **471**, a grinding block **474** used for grinding is mounted on an outer side of a middle section of the driven shaft **471**, a grinding ring **475** is disposed on an outer side of the grinding block **474**, foam particles can be ground by means of the grinding block **474** and the grinding ring **475**, the grinding ring **475** is mounted on an inner wall of a lantern ring **476** through bolts, an outer side of the lantern ring **476** is fixedly mounted on an inner circular surface of the spraying plate **4661** through a heat-insulation cross rod **477**, opposite upper and lower sides of the grinding block **474** and the grinding ring **475** are arc-shaped to reduce the flowing resistance of a foam material, and the distance between the grinding block **474** and the grinding ring **475** is 1 cm.

(19) The working principle of the supercritical in-mold foam forming device with a pretreatment function and the foaming method thereof is as follows:

(20) Step 1: when used, the device is placed in a working area and is then connected to an external power supply, such that power is supplied to the device for work;

(21) Step 2: an external pipe is connected to the one-way metering valve **42** by a worker, and an inert gas is guided into the heating box **41** to be heated, wherein the inert gas is specifically carbon dioxide commonly used during a foaming process; then, the heated inert gas is guided into the radial gas cutting module **46** through the temperature control valve **45**;

(22) Step 3: the gas flow is transported to the hollow ring **464** by means of the diverter valve **461** through the rigid tubes **463**, such that the gas flow passes through the hollow ring **464** and then enters the ceramic plate **465**; at this moment, the pressure in the ceramic plate **465** is increased to push the sealing plug **4666** to squeeze the elastic element **4665**, such that the sealing plug **4666** horizontally moves in the through groove **4662**;

(23) Step 4: when the sealing plug **4666** moves to make a strip-shaped groove in an outer side of the sealing plug **4666** gradually separated from a strip-shaped protrusion of the through groove **4662**, the gas flow enters the conical assembly **4663** through a gap and is guided into the high-pressure reactor **1** through the polymer heat-resistant membrane **4664** to sweep and preheat the interior of the high-pressure reactor **1**;

(24) Step 5: then, the inert gas flows back into the heating box **41** through a pipe at the top of the high-pressure reactor **1**, and the safety valve **2** is opened to add a foam material and a foaming agent into the high-pressure reactor **1**;

(25) Step 6: then, the driving member **3** is controlled through the control panel **6** to drive the planetary gear set **472** to drive the driven shaft **471** and the hollow ring **464** to rotate in opposite directions, at this moment, the driven shaft **471** synchronously drives the axial blades **473** and the grinding block **474** to rotate, and the axial blades **473** drive the foam material and the foaming agent to flow axially in a direction parallel to the driven shaft **471**, such that the foam material and the foaming agent flow from bottom to top to enter the hollow ring **464**, and the grinding block **474** and the grinding ring **475** rotate in opposite directions to grind the foam material (the problems that, in the prior art, the grinding ring **475** is fixedly arranged and only the grinding block **474** can rotate, so the grinding speed is low and the grinding effect is unsatisfactory are solved); then, the inert gas is guided into the radial gas cutting module **46** and is sprayed out through the through hole in the inner side of the through groove **4662**;

(26) Step 7: the foaming agent is prevented from flowing into the through groove **4662** under the action of the polymer heat-resistant membrane **4664** and the gas pressure, such that the foam material and the foaming agent are sheared and stirred in a radial direction by the radial gas cutting module **46**, and are stirred and ground in an axial direction by the axial blades **473** and the grinding block **474**, such that the foaming agent infiltrates into the foam material more quickly;

(27) Step 8: after the foaming agent infiltrates into the foam material, the two groups of uniform

heating plates **442** are controlled by means of the programmable control box **43** to be powered on synchronously to generate heat, and at this moment, power output to the two groups of uniform heating plates **442** is controlled to be different, wherein the input power of the uniform heating plate **442** at the top is maximum, and the input power of the uniform heating plates **442** below is decreased in proportion; the difference in output heat of the uniform heating plates **442** leads to a temperature gradient in the graphite backing plate **444**, the heat is reflected around the high-sensitivity temperature gauge **445** in equal proportion through filler materials in the intermediate mold body **443**, and the proportion of the input power of the uniform heating plates **442** is calculated by adjusting a heat value gradient detected by the high-sensitivity temperature gauge **445** according to a preset value in the programmable control box **43**;

(28) Step 9: a foamed preform is discharged out of the foaming mold **5** through a discharge unit which is a common device used during the foaming process such as an extruder, then the foamed preform enters the graphite backing plate **444**, the 3D foaming ratio of the 3D preform is nonsynchronous under the influence of the temperature gradient, with the foaming ratio along the length and width in the horizontal direction being small and the foaming ratio in the vertical direction being large, and then compression molding is performed, wherein to guarantee the nonsynchronous 3D foaming ratio, the thickness of an internal space of the graphite backing plate **444** is kept at a value which ensures that the heat output efficiency of the uniform heating plates **442**, the heat conductivity of the graphite backing plate **444** and the foam material, and the thickness of the internal space of the graphite backing plate are in direct proportion.

(29) According to the supercritical in-mold foam forming device with a pretreatment function and the foaming method thereof provided by the invention through improvement. the pretreatment module **4** is disposed at the front end of the high-pressure reactor **1**, an inert gas is guided, through the one-way metering valve **42**, into the heating box **41** to be heated, then the heated inert gas is guided into the radial cutting module **46** through the temperature control valve **45**, the axial stirring module **47** is driven by the driving member **3**, and the foam material and the foaming agent are sheared and stirred in radial direction by the radial gas cutting module **46** and are stirred and ground in the axial direction by the axial stirring module **47**, such that the foaming agent can infiltrate into the foam material more quickly; the intra-mold temperature control module **44** is disposed in the foaming mold **5**, the two groups of uniform heating plates **442** are controlled to be powered on synchronously to generate heat, a temperature gradient is formed in the graphite backing plate **444** by controlling the difference in proportion of output power, heat is reflected around the high-sensitivity temperature gauge **445** in equal proportion through the filler materials in the intermediate mold body **443**, the foamed preform is conveyed into the graphite backing plate **444**, the 3D foaming ratio of the 3D preform is nonsynchronous under the influence of the temperature gradient, with the foaming ratio along the length and width in the horizontal direction being small and the foaming ratio in the vertical direction being large, and then compression molding is performed; the radial gas cutting module **46** is disposed in the high-pressure reactor **1**, the gas flow from the temperature control valve **45** enters the ceramic plate **465** through the rigid tubes **463** and the hollow ring **464** by means of the diverter valve **461**, and at this moment, the gas pressure in the ceramic plate **465** is increased to push the sealing plug **4666** to squeeze the elastic element **4665**, such that the scaling plug **4666** horizontally moves in the through groove **4662**; at this moment, the gas flow enters the conical assembly **4663** through the gap and is guided into the high-pressure reactor **1** through the polymer heat-resistant membrane **4664**, the sealing plug **4666** is pushed by the gas pressure to move, the through groove **4662** is communicated with the interior of the ceramic plate **465** through the strip-shaped groove formed in the outer side of the sealing plug **4666**, the space between the through groove **4662** and the interior of the ceramic plate **465** can be quickly sealed by means of the elastic potential energy of the elastic element **4665** when the gas pressure decreases, and the foaming agent is prevented from flowing into the through groove **4662** by means of the polymer heat-resistant membrane **4664** which is made of a water-proof and

breathable material, so the hollow ring **464** is prevented from being blocked, the situation where the proportion of materials is out of balance due to the loss of the foaming agent is avoided, the influence on foaming infiltration is reduced, and the interior of the high-pressure reactor **1** is better swept and preheated; the axial stirring module **47** is disposed in the high-pressure reactor **1**, the driving member **3** drives the planetary gear set **472** to drive the driven shaft **471** and the hollow ring **464** to rotate in opposite directions, and at this moment, the axial blades **473** drive the foam material and the foaming agent to flow axially in a direction parallel to the driven shaft **471**, such that the foam material and the foaming agent flow from bottom to top to enter the hollow ring **464**; the grinding block **474** and the grinding ring **475** rotate in opposite directions to grind the foam material, and then an inert gas is guided into the radial gas cutting assembly **46** to shear and stir the foam material and the foaming agent in the radial direction.

(30) The basic principle, main features and advantages of the invention are illustrated and described above, standard parts used in the invention can be purchased from the market, special-shaped parts can be customized according to the specification and drawings, all parts are connected specifically through mature and common methods in the prior art such as bolt riveting or welding, all machines, parts and equipment are of common models in the prior art, common connection methods in the prior art are used for circuit connection, and details will not be given here.

(31) With reference to the description of the embodiments disclosed above, those skilled in the art can implement or use the invention. Various modifications to these embodiments will be obvious to those skilled in the art. The general principle defined in this specification can be implemented in other embodiments without departing from the spirit or scope of the invention. Thus, the invention should not be limited to the embodiments in this specification, and should have the broadest scope in conformity with the principle and novel features disclosed in this specification.

Claims

1. A supercritical in-mold foam forming device with a pretreatment function, comprising a high-pressure reactor (**1**), a safety valve (**2**) disposed at a top of the high-pressure reactor (**1**) and sealing the top of the high-pressure reactor (**1**), a driving member (**3**) mounted by bolts in a middle of the top of the high-pressure reactor (**1**), a foaming mold (**5**) mounted by bolts on a support frame at a bottom of the high-pressure reactor (**1**), and a control panel (**6**) mounted by bolts in a middle of a right end of the high-pressure reactor (**1**); wherein the supercritical in-mold foam forming device further comprises a pretreatment module (**4**) disposed at a front end of the high-pressure reactor (**1**); the pretreatment module (**4**) comprises: a heating box (**41**) for heating an inert gas to control the temperature of the inert gas and mounted by bolts at a front end of the high-pressure reactor (**1**); a one-way metering valve (**42**) fixedly mounted at a pipe opening connected to a left end of the heating box (**41**); a programmable control box (**43**) mounted by bolts at a top of the heating box (**41**), a heat-insulation cushion being disposed between the heating box (**41**) and the programmable control box (**43**); an intra-mold temperature control module (**44**) disposed in the foaming mold (**5**) and connected to the programmable control box (**43**) by a cable; a temperature control valve (**45**) mounted at a through hole in a bottom of a right end of the heating box (**41**) through a pipe; a radial gas cutting module (**46**) disposed in the high-pressure reactor (**1**); and an axial stirring module (**47**) disposed at a bottom of the driving member (**3**); wherein the driving member (**3**) is composed of a gearbox and a motor, and the one-way metering valve (**42**) and the temperature control valve (**45**) are connected to interfaces of the programmable control box (**43**) by respective cables.
2. The supercritical in-mold foam forming device with a pretreatment function according to claim 1, wherein the intra-mold temperature control module (**44**) comprises: matching mold bodies (**441**) fixedly mounted on opposite upper and lower sides of the foaming mold (**5**) and in seal fit with the upper and lower sides of the foaming mold (**5**); uniform heating plates (**442**) mounted by bolts on inner sides of the matching mold bodies (**441**) on the upper and lower sides of the foaming mold

(5); an intermediate mold body (443), the matching mold bodies (441) being in seal fit with upper and lower sides of the intermediate mold body (443); a graphite backing plate (444) for heat conduction and disposed on an inner wall of the intermediate mold body (443); a high-sensitivity temperature gauge (445) disposed in the intermediate mold body (443); a data switching unit (446) connected to and mounted at a bottom of the high-sensitivity temperature gauge (445) by a cable; and a horizontal connecting plate (447) inserted and fixed in a front side of the intermediate mold body (443).

3. The supercritical in-mold foam forming device with a pretreatment function according to claim 2, wherein a heat-conducting material layer is filled and fixed in the inner wall of the intermediate mold body (443), and the high-sensitivity temperature gauge (445) is fixed in an internal space of the heat-conducting material layer.

4. The supercritical in-mold foam forming device with a pretreatment function according to claim 1, wherein the radial gas cutting module (46) comprises: a diverter valve (461), a right end of the temperature control valve (45) being connected to a gas port in a bottom of the diverter valve (461) through a pipe; a retainer ring (462) rotatably disposed at a top of the diverter valve (461); rigid tubes (463) for flow guiding and inserted and fixed in through holes in an outer circular side of the diverter valve (461); a hollow ring (464), the rigid tubes (463) being connected to through holes in an inner circular wall of the hollow ring (464) through pipes; a ceramic plate (465), the hollow ring (464) being connected to upper and lower sides of the ceramic plate (465) by bolts respectively; a sprayer (466) disposed on an inner circular wall of the ceramic plate (465); and heat-resistant ring tubes (467), through holes in the upper and lower sides of the ceramic plate (465) being connected to a gas outlet of the hollow ring (464) through heat-resistant ring tubes (467) respectively.

5. The supercritical in-mold foam forming device with a pretreatment function according to claim 4, wherein a central point of the retainer ring (462) and a center of the hollow ring (464) are located at a same height, six groups of said rigid tubes (463) are disposed on the outer side of the diverter valve (461), and outer sides of the rigid tubes (463) are coated with heat-insulation coatings.

6. The supercritical in-mold foam forming device with a pretreatment function according to claim 4, wherein the sprayer (466) comprises: a spraying plate (4661) mounted by bolts on an inner wall of the ceramic plate (465); a through groove (4662) formed in the spraying plate (4661); a conical assembly (4663) fixedly mounted in the through groove (4662); a polymer heat-resistant membrane (4664) filled in a space between the conical assembly (4663) and the through groove (4662); an elastic element (4665) disposed on a conical head of the conical assembly (4663); and a sealing plug (4666), an outer end of the elastic element (4665) being elastically connected to the sealing plug (4666).

7. The supercritical in-mold foam forming device with a pretreatment function according to claim 6, wherein a gas hole is formed in a position, communicated with the through groove (4662), of the conical assembly (4663), the through groove (4662) and the conical assembly (4663) are identical in shape and are distributed in a proportion of 1:1.2, strip-shaped gas holes and strip-shaped lugs matched with the strip-shaped gas holes are disposed on the sealing plug (4666) and a through hole in a rear end of the through groove (4662), and the polymer heat-resistant membrane (4664) is a water-proof and breathable membrane.

8. The supercritical in-mold foam forming device with a pretreatment function according to claim 6, wherein the axial stirring module (47) comprises: a driven shaft (471) inserted and fixed in an inner side of the retainer ring (462); a planetary gear set (472), a top of the driven shaft (471) being inserted and fixed in a center of a sun gear of the planetary gear set (472), the planetary gear set (472) being composed of the sun gear, a planetary gear and a gear ring, the planetary gear of the planetary gear set (472) being inserted and fixed in a shaft of the gearbox of the driving member (3), and a gear ring plate of the planetary gear set (472) being fixed to an inner wall of the hollow ring (464); axial blades (473) inserted and fixed to upper and lower sides of an outer shaft of the driven shaft (471); a grinding block (474) for grinding and mounted on an outer side of a middle

section of the driven shaft (471); a grinding ring (475) disposed on an outer side of the grinding block (474); a lantern ring (476), the grinding ring (475) being mounted by bolts on an inner wall of the lantern ring (476); and a heat-insulation cross rod (477), an outer side of the lantern ring (476) being fixedly mounted on an inner circular surface of the spraying plate (4661) through the heat-insulation cross rod (477).

9. The supercritical in-mold foam forming device with a pretreatment function according to claim 8, wherein opposite upper and lower sides of the grinding block (474) and the grinding ring (475) are arc-shaped, and a distance between the grinding block (474) and the grinding ring (475) is 1 cm.

10. A foaming method of the supercritical in-mold foam forming device with a pretreatment function according to claim 9, comprising: Step 1: first, connecting, by a worker, an external pipe to the one-way metering valve (42), and guiding an inert gas into the heating box (41) to be heated, wherein the inert gas is carbon dioxide; then, guiding the heated inert gas into the radial gas cutting module (46) through the temperature control valve (45) to sweep and preheat an interior of the high-pressure reactor (1), then enabling the inert gas to flow back into the heating box (41) through a pipe at the top of the high-pressure reactor (1), and opening the safety valve (2) to add a foam material and a foaming agent into the high-pressure reactor (1); Step 2: controlling, by the control panel (6), the driving member (3) to drive the planetary gear set (472) to drive the driven shaft (471) and the hollow ring (464) to rotate in opposite directions while driving, by the axial blades (473), the foam material and the foaming agent to flow upward so as to enter the hollow ring (464), causing the grinding block (474) and the grinding ring (475) to rotate in opposite directions to grind the foam material, and then guiding the inert gas into the radial gas cutting module (46) so as to spray out through the through groove (4662) such that the foam material and the foaming agent are sheared and stirred in a radial direction by the radial gas cutting module (46), and are stirred and ground in an axial direction by the axial blades (473) and the grinding block (474), thereby causing the foaming agent to infiltrate into the foam material quickly; Step 3: controlling, by means of the programmable control box (43), the uniform heating plates (442) to be powered on synchronously to generate heat while controlling power output to the uniform heating plates (442) to be different, wherein input power of the uniform heating plate (442) at the upper side of the foaming mold (5) is maximum, and input power of the uniform heating plate (442) at the lower side of the foaming mold (5) is decreased relative thereto; a difference in output heat of the uniform heating plates (442) leading to a temperature gradient in the graphite backing plate (444), the heat being reflected around the high-sensitivity temperature gauge (445) in equal proportion through filler materials in the intermediate mold body (443), and the proportion of the input power of the uniform heating plates (442) is calculated by adjusting a heat value gradient detected by the high-sensitivity temperature gauge (445) according to a preset value in the programmable control box (43), thereby forming a foamed preform in the foaming mold (5); and Step 4: discharging the foamed preform out of the foaming mold (5) through a discharge unit and causing the foamed preform to enter the graphite backing plate (444), making a 3D foaming ratio of the preform nonsynchronous under the influence of the temperature gradient, with a foaming ratio along a length and width in a horizontal direction of the foamed preform being small and a foaming ratio in a vertical direction of the foamed preform being large, and then performing compression molding of the foamed preform.
